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(54) **CRYSTALLINE FORMS OF ERAVACYCLINE**

KRISTALLINE FORMEN VON ERAVACYCLIN

FORMES CRISTALLINES DE L'ÉRAVACYCLINE

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• **MAGNUS RONN ET AL: "Process R&D of Eravacycline: The First Fully Synthetic Fluorocycline in Clinical Development", ORGANIC PROCESS RESEARCH AND DEVELOPMENT, vol. 17, no. 5, 17 May 2013 (2013-05-17), pages 838-845, XP055254495, US ISSN: 1083-6160, DOI: 10.1021/op4000219**
• **CAIRA ED - MONTCHAMP JEAN-LUC: "Crystalline Polymorphism of Organic Compounds", TOPICS IN CURRENT CHEMISTRY; [TOPICS IN CURRENT CHEMISTRY], SPRINGER, BERLIN, DE, vol. 198, 1 January 1998 (1998-01-01), pages 163-208, XP008166276, ISSN: 0340-1022**

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Description

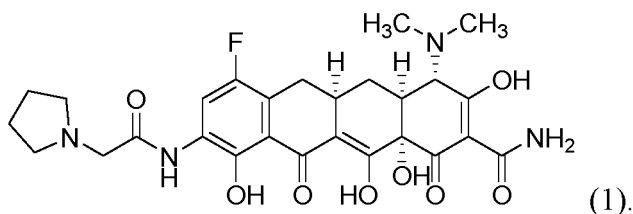
GOVERNMENT SUPPORT

[0001] This invention was made with government support under Contract No. HHSO100201200002C and Subcontract No. 7834S1 awarded by the Department of Health and Human Services. The government has certain rights in the invention.

BACKGROUND OF THE INVENTION

[0002] The tetracyclines are broad spectrum anti-microbial agents that are widely used in human and veterinary medicine. The total production of tetracyclines by fermentation or semi-synthesis is measured in the thousands of metric tons per year.

[0003] The widespread use of tetracyclines for therapeutic purposes has led to the emergence of resistance to these antibiotics, even among highly susceptible bacterial species. Tetracycline analogs having improved antibacterial activities and efficacies against other tetracycline responsive diseases or disorders have been described (see, for example, US Patent No. 8,796,245). A particularly potent compound is eravacycline (7-fluoro-9-pyrrolidinoacetamido-6-demethyl-6-deoxytetracycline), which has the chemical structure shown in Structural Formula 1:



[0004] An article by Ronn, M. et al. entitled "Process R&D of Eravacycline: The First Fully Synthetic Fluorocycline in Clinical Development" was published in Organic Process Research and Development, vol. 17, no. 5 (17 May 2013), pages 838-845. WO 2016/065290 A1, entitled "Semi-synthesis procedures", was published on 28 April 2016.

[0005] The solid form of a compound can be important in the formulation of pharmaceutical compositions. For example, crystalline and amorphous forms of a compound can have different physical properties (e.g., stability, dissolution rate, density, etc.) relating to their suitability for use in pharmaceutical compositions. The difference in physical properties can also affect a crystalline or amorphous form's usefulness, for example, as an intermediate in the synthesis of a form suitable for use in pharmaceutical compositions.

[0006] There is a need for crystalline forms of eravacycline that are thermodynamically stable and suitable for use in pharmaceutical compositions (e.g., are readily dissolvable, exhibit good flow properties and/or good chemical stability). There is a further need for crystalline forms of eravacycline having physical properties that enable the manufacture of eravacycline for use in pharmaceutical compositions in high yield and high purity.

SUMMARY OF THE INVENTION

[0007] The present invention relates to crystalline forms of the bis HCl salt of the compound represented by Structural Formula 1, designated crystalline Form I and crystalline Form J and compositions comprising the crystalline forms. Crystalline form I is a hemi-ethanolate solvate.

[0008] In one embodiment, a crystalline form of a compound represented by the bis HCl salt of Structural Formula 1 is provided, wherein the crystalline form is Form I. In this embodiment, crystalline Form I is characterized by at least three x-ray powder diffraction peaks at 2θ angles selected from 7.22° , 7.80° , 10.41° , and 11.11° .

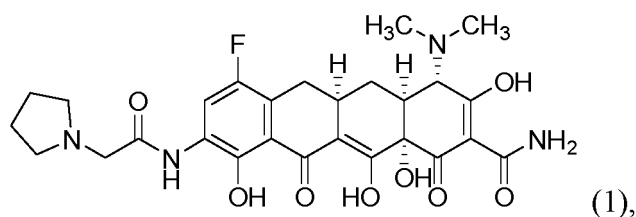
[0009] In another embodiment, a crystalline form of a compound represented by bis HCl salt of Structural Formula 1 is provided, wherein the crystalline form is Form J. In this embodiment, crystalline Form J is characterized by at least three x-ray powder diffraction peaks at 2θ angles selected from 7.02° , 7.80° , 22.13° , and 23.22° .

[0010] It is also disclosed a crystalline form of a compound represented by the bis HCl salt of Structural Formula 1 is provided, wherein the crystalline form is Form A. In this embodiment, crystalline Form A is characterized by at least three x-ray powder diffraction peaks at 2θ angles selected from 3.31° , 6.01° , 6.33° , and 8.73° .

[0011] It is also disclosed crystalline form of a compound represented by the bis HCl salt of Structural Formula 1 is provided, wherein the crystalline form is Form B. In this embodiment, crystalline Form B is characterized by at least three x-ray powder diffraction peaks at 2θ angles selected from 9.19° , 9.66° , 23.32° , and 24.35° .

[0012] Another embodiment, is a composition comprising particles of one or more crystalline forms of a compound

represented by the bis HCl salt of Structural Formula 1:



wherein the particles are selected from: particles of crystalline Form I characterized by at least three x-ray powder diffraction peaks at 2θ angles selected from 7.22° , 7.80° , 10.41° , and 11.11° ; particles of crystalline form Form J characterized by at least three x-ray powder diffraction peaks at 2θ angles selected from 7.02° , 7.80° , 22.13° , and 23.22° ; are also disclosed particles of crystalline Form A characterized by at least three x-ray powder diffraction peaks at 2θ angles selected from 3.31° , 6.01° , 6.33° , and 8.73° ; and particles of crystalline Form B characterized by at least three x-ray powder diffraction peaks at 2θ angles selected from 9.19° , 9.66° , 23.32° , and 24.35° . In a specific aspect, the composition comprises particles of crystalline Form I and crystalline form J. In another aspect, the weight percent of crystalline Form J in the composition is 25% or less (e.g., about 20% or less, about 15% or less, about 10% or less, about 5% or less or about 1%).

[0013] Another embodiment is a pharmaceutical composition comprising particles of Form I, Form J, or mixtures thereof and a pharmaceutically acceptable carrier

[0014] A further embodiment is a crystalline form I or J of the compound of formula I for use in a method for treating or preventing a tetracycline-responsive disease or disorder, the method comprising administering to a subject in need thereof a therapeutically or prophylactically effective amount of crystalline Form I, Form J, or a mixture thereof. In one aspect, the tetracycline-responsive disease or disorder is an infection. In a specific aspect, the infection is caused by bacteria.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The foregoing will be apparent from the following more particular description of example embodiments of the invention.

FIG. 1 is an x-ray powder diffraction (XRPD) pattern of Structural Formula 1 as the bis HCl salt)- Form A.

FIG. 2A is a thermogravimetric analysis (TGA) thermogram of Structural Formula 1 - Form A.

FIG. 2B is a differential scanning calorimetry (DSC) thermogram of Structural Formula 1 - Form A.

FIG. 3 is an XRPD pattern of Structural Formula 1 - Form B.

FIG. 4A is a TGA thermogram of Structural Formula 1 - Form B.

FIG. 4B is a DSC thermogram of Structural Formula 1 - Form B.

FIG. 5 is an XRPD pattern of Structural Formula 1 - Form I.

FIG. 6A is a TGA thermogram of Structural Formula 1 - Form I.

FIG. 6B is a DSC thermogram of Structural Formula 1 - Form I

FIG. 7 is an XRPD pattern of Structural Formula 1 - Form J.

FIG. 8A is a TGA thermogram of Structural Formula 1 - Form J.

FIG. 8B is a DSC thermogram of Structural Formula 1 - Form J.

FIG. 9 is a DVS of Structural Formula 1-Amorphous Form

FIG. 10 is a DVS of Structural Formula 1-Form A

FIG. 11 is a DVS of Structural Formula 1-Form B

FIG. 12 is a DVS of Structural Formula 1-Form I

FIG. 13 is a DVS of Structural Form 1-Form J

DETAILED DESCRIPTION OF THE INVENTION

[0016] A description of example embodiments of the invention follows.

Crystalline Forms of Eravacycline

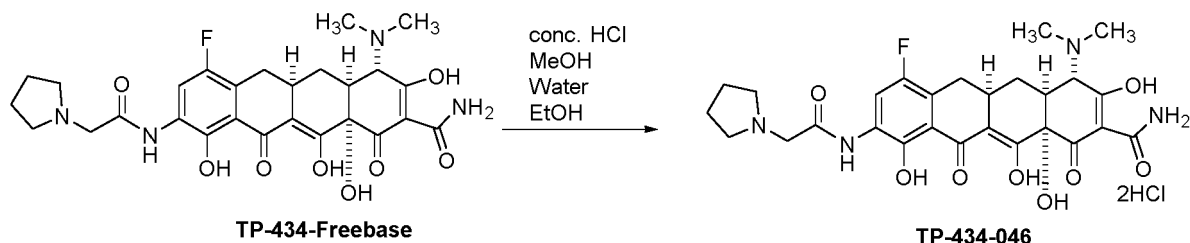
[0017] Provided herein are crystalline forms of the bis HCl salt of the compound represented by Structural Formula 1, designated crystalline Form A, crystalline Form B, crystalline Form I and crystalline Form J.

Example 2 - Form B

Preparation of Form B without seeding

Conversion of TP-434-Freebase to TP-434-046 Form B

[0160]



Procedure:

[0161] To a 5 L dried flask was charged 814 ml anhydrous methanol, followed by 358 mL 200 proof anhydrous ethanol. The mixture was cooled to 0 to 2°C. Water (81 mL) was charged followed by 199.7 ml 37% concentrated HCl. Temperature rose from 1.8°C to 19.0°C. After temperature of the reaction mixture returned to 0°C, TP-434 freebase (426 g, about 0.763 moles,) was charged as a solid slowly over 30 minutes with temperature maintained below 20°C.

[0162] Solid suspension appears after only 4 minutes and microscope shows plate like crystal (Form B) formed. The yellow suspension was stirred for 2 hours at 20 to 23°C.

[0163] Anhydrous 200 proof ethanol (1.91L) was added over 3 hours into the mixture. The resulted yellow suspension was cooled to 0 to 1°C over 1 hour and the suspension was continuously stirred at 0 to 2°C for 30 minutes.

[0164] The solid was collected by filtration through "M" fritted filter funnel. The solid cake was washed with 250 ml anhydrous 200 proof ethanol. After the wash, the cake was continuously dried in the filter funnel by pulling vacuum with nitrogen blanket for 14 hours.

[0165] The solid (610 g) was transferred into two trays and dry in high vacuum oven at room temperature (25±5°C) until achieving constant weight (took about 24 hours) to give a yellow solid (466 g). KF is 6.2%. ¹H-NMR showed the molar ratio of TP-434-046/MeOH/EtOH was 1/0.97/0.16.

[0166] The solid (459g) was further dried under vacuum with humidity (with a beaker of water in the drying oven, 60-80% humidity, vacuum 0.043 mpa) for one day and then dried under vacuum by oil pump for two days to afford TP-434-046 as light yellow solid (458 g).

[0167] HPLC of product showed 98.88% TP-434. Assay of w/w showed the product contained 75.3% TP-434 Freebase. (Based on the w/w assay of the product, the yield of the crystallization is as least 81% without w/w assay correction of the starting material.) The residual methanol was 1.3 wt% and ethanol was not detected by ¹H-NMR. The residue water by KF was 11.18%. CHN and chloride analysis showed C, 45.07 %; H, 5.80 %; N, 7.88 %; Cl-, 9.88 %. Calc. for C₂₇H₃₃Cl₂F₁N₄O₈·5H₂O: C, 44.94 %; H, 6.01 %; N, 7.76 %; Cl-, 9.83; water, 12.5%. The XRPD of product was consistent with Form B.

Preparation of Form B with seeding

[0168] Crude TP-434-046 was dissolved in methanol (6V). The solution was polish filtered through a 0.45μm inline filter and rinsed with a solution of methanol (1V), water (0.56V) and concentrated HCl (0.16V). The solution was seeded with Form B seeds (0.5wt%) prepared as described above and stirred at 21±2°C for approximately 4 hours. Polished filtered concentrated HCl (0.28V) was added slowly with temperature controlled at 21±2°C. The mixture was stirred at 21±2°C for about 16 hours, cooled to 0°C in about 30 minutes and held at 0°C for 3 hours. The solid was collected by filtration, washed with a solution of MeOH/water/concentrated HCl (2×1V, 0.87/0.07/0.06). The solid (Form B) was dried under both vacuum and nitrogen.

[0169] Form B was characterized as a methanolate (with 1eq of methanol) with plate shaped crystals under microscope and distinctive XRPD. Crystallization of Form B provide high recovery yield and good ability for removing impurities. While Form B may not be the most desired form for pharmaceutical use due to the presence of methanol, the desirable purity and high yield of crystalline Form B from amorphous TP-434-046 can be translated into pure crystalline Form I, Form J or a mixture thereof for pharmaceutical use.

[0170] Form B is a co-solvate with MeOH (about 1eq) and water.

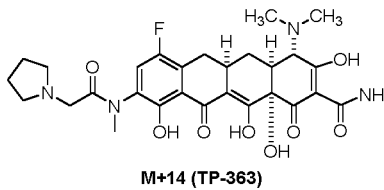
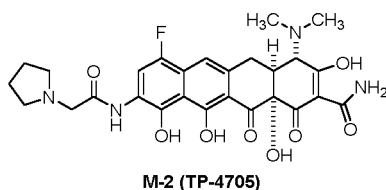
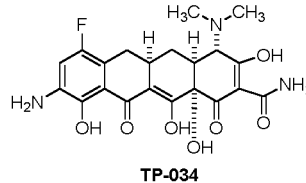
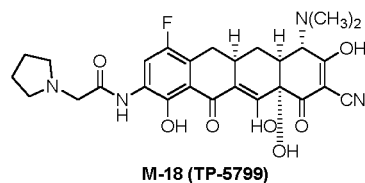
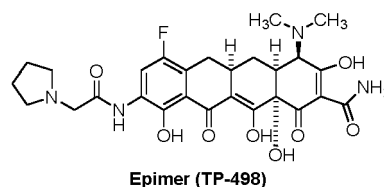
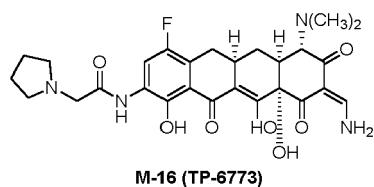
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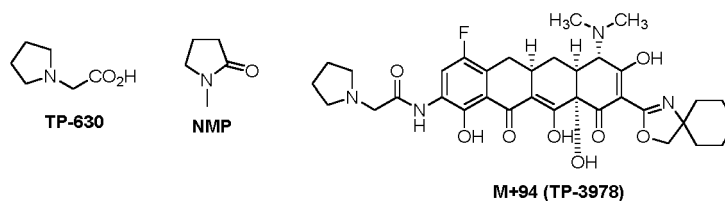
the necessary levels by Form B crystallization (see, e.g., NMP and TP-630). While Form B may not be the most desirable Form described herein to use as an API because it is a methanolate, the high impurity purging ability is advantageous for conversion to thermodynamically stable Form I, Form J or a mixture thereof.

Table 8:

Impurity Purging (Reduction of Impurity Level)				
Impurity	Amorphous Form Precipitation	Form A crystallization	Form B crystallization	Slurry Form B to Form I/J
M-16 (TP-6773)	No purging	5-8%	25-28%	Increased $\approx$ 10%
Epimer (TP-498)	No purging	Up to 40%	70%	40-80%
M-18 (TP-5799)	No purging	90%	30-60%	25-40%
TP-034	No purging	90%	50-79%	50-80%
M-2 (TP-4705)	NA	NA	50%	90%
M+14 (TP-363)	NA	>95%	62%	85%
TP-630	54%	>95%	>99%	>99%
NMP	97%	>95%	>99%	>99%
M+94 (TP-3978)	NA	NA	>98%	61%

The structures of the impurities are set forth below:



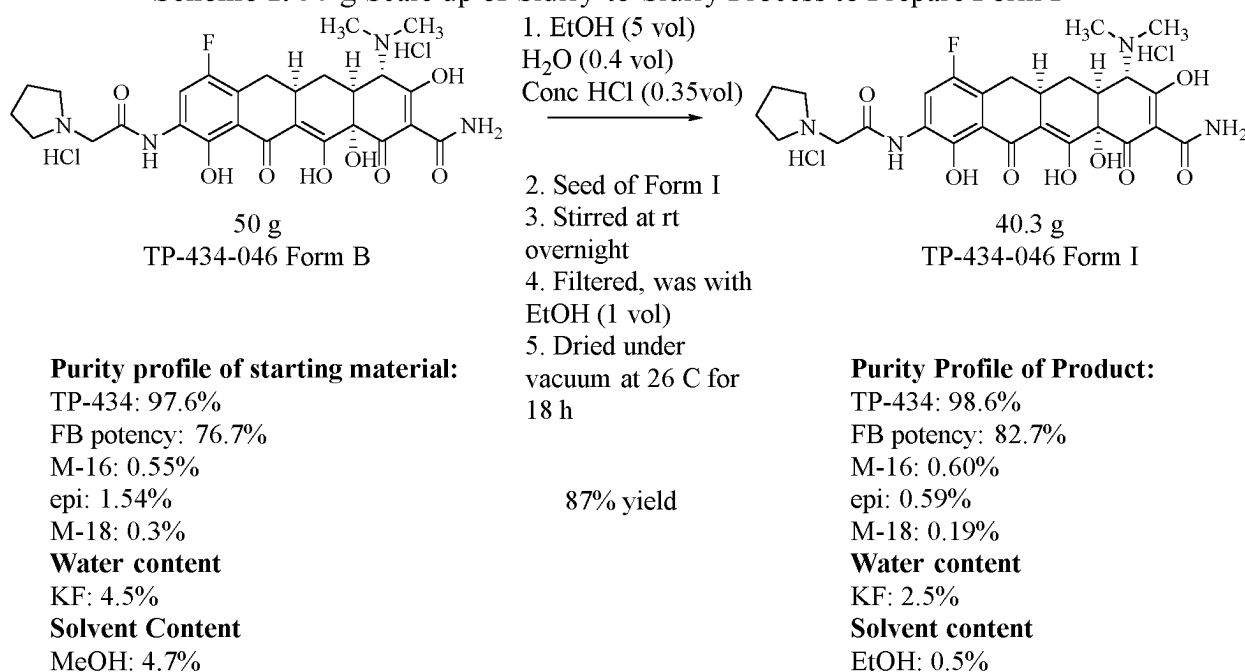


Example 11 - Large Scale Batches of Form I and Form J

50-g scale-up batch

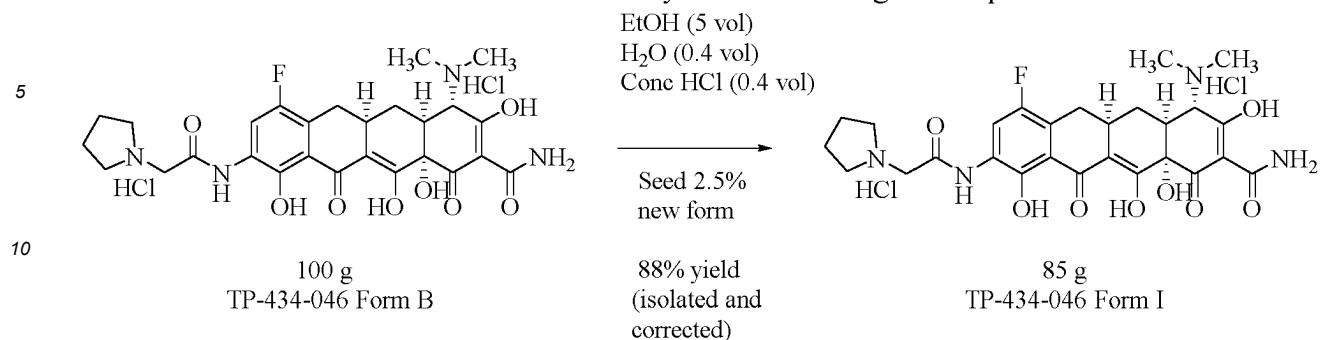
[0199] A 50-g scale-up of the slurry-to-slurry process was performed. Form B was converted to the new form overnight. The supernatant concentration after 18 hours was 21 mg/g (TP-434 free base). The solids were filtered (fast filtration), washed with ethanol (1 vol) and dried in vacuum oven at 26 °C. Scheme 1 summarizes the results from this experiment. XRPD pattern of this material confirmed that it was mostly Form I with minor Form J. The freebase (FB) potency of TP-434 was determined by HPLC weight/weight assay using TP-434 reference standard.

Scheme 1. 50-g Scale up of Slurry-to-Slurry Process to Prepare Form I



100-g scale-up batch

[0200] Preparation of Form I was scaled up using 100 g of Form B. The experiment was carried out by adding Form B into a solution of 5 vol of EtOH, 0.4 vol of water, 0.4 of conc. HCl followed by charging Form I seeds (2.5 wt% seed loading) and a total of 85 g of Form I was isolated. Scheme 2 summarizes the data. The purity of Form I was consistent with previous batches.

Scheme 2. Form B slurry to Form I 100-g scale-up**Purity Profile of starting Material:**

TP-434: 97.6%
 FB potency: 76.7%
 M-16: 0.55%
 epi: 1.54%
 M-18: 0.3%

Water Content

KF: 4.47%

Solvent content

MeOH: 4.66%

Purity Profile of Product:

TP-434: 98.58%
 FB potency: 81.0%
 M-16: 0.61%
 epi: 0.61%
 M-18: 0.2%

Water Content

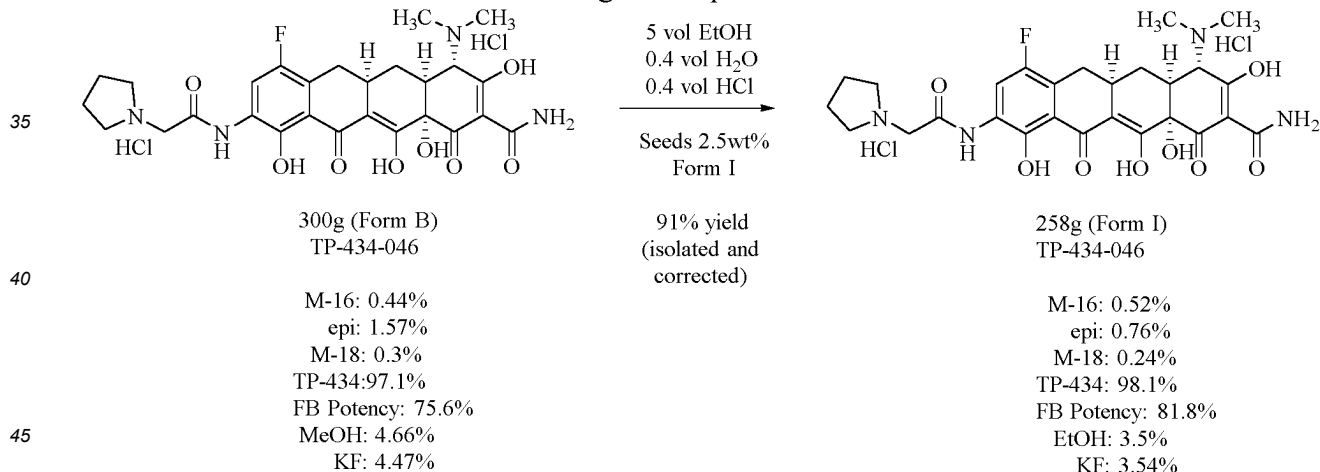
KF: 4.2%

Solvent content

EtOH: 3.3%

300-g scale-up batch

[0201] A 300-g scale slurry to slurry conversion from Form B to Form I was performed. The batch overview is presented in Scheme 3:

Scheme 3. 300-g scale up of Form I Process

[0202] The rate of the 300-g scale crystallization was slower than expected. After overnight stirring at room temperature, the supernatant concentration for Form I crystallization usually stands at 8-15 mg/mL of free base. For the 300-g scale batch however, this concentration was 39 mg/mg. Cooling the suspension to 10 °C, then 0 °C, did not lower the concentration as fast as the equivalent operation for Form A. After 30 hours overall time, 1 volume of EtOH was added to the suspension at 0 °C. The batch was allowed to stir at room temperature up to 48 hours total time. At this point, the concentration reached 11 mg/mL and the batch was filtered, washed and the product dried in vacuum oven overnight to afford TP-434-046 (Form I)

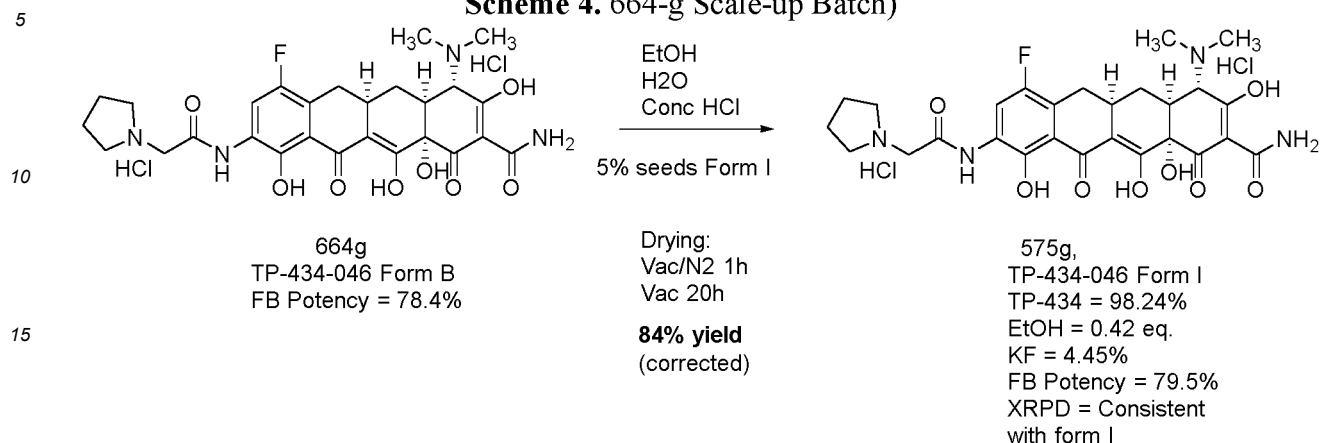
664-g scale-up batch

[0203] Starting with Form B (664 g), the scale-up was carried out using the current Form B to Form I conversion

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procedure to afford TP-434-046 (575 g, 84% yield, containing 3.1% EtOH and 4.5% water) after drying in a vacuum oven. The batch overview is shown in Scheme 4. The XRPD was consistent with Form I with some Form J.

Scheme 4. 664-g Scale-up Batch)



1.83-kg scale-up batch

[0204] A Scale-up batch was made starting with 1.83kg of Form B. The detailed procedure is described in the followings.

Scheme 5. 1.83-kg batch

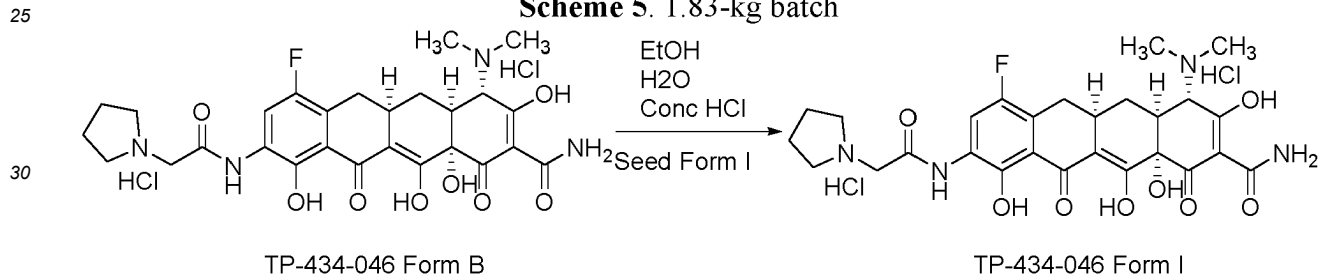


Table 9

Material	Qty	MW	Mole eq.
TP-434-046 form B	1830 g	631.5	1.0
EtOH 200 proof	9150 mL	-	5V
Water-HPLC grade	732 mL	-	0.4V
Conc HCl	732 mL	-	0.4V
TP-434-046 form I seeds	90 g	631.5	5 % wt
EtOH 200 proof	5490 mL	-	3V
EtOH 200 proof	3294 mL	-	1.8 V
Water	183 mL	-	0.1 V
Conc HCl	183 mL	-	0.1 V

1. To a 20 L reactor, charge EtOH (9150 mL), H₂O (732 mL) and concentrated HCl (732 mL). **Note:** Filtered the solution through the 0.22- μ m filter prior to use.

2. Charge TP-434-046 form B (1830 g) in one portion and stir for 5 minutes to allow the solution to reach saturation.

3. Seed with form I (90g) in one portion and stir at room temperature overnight 18 hours.

4. Check the Microscope, form B convert into Form I. **Note:** Confirm FORM I

5. Supernatant wt/wt assay is 1.77% using HPLC.

shoulder of the first peak of Form I around $7^\circ 2\theta$. Batch 3 contains more Form J, as demonstrated by the peak on the shoulder of the first peak of Form I around $7^\circ 2\theta$. All three registration batches met the XRPD spec defined as "consistent to Form I, Form J, or a mixture of Forms I and J."

[0212] With respect to Batch 1, which contains pure Form I, the formation of Form A as transient kinetic form during the Form B to Form I conversion may have affected the product XRPD and particle size. The particles in Batch 1 are mostly composed of smaller well-defined individual crystals, while the particles in Batch 2 and Batch 3 are composed with large crystals containing clusters of individual crystals. The morphologies of crystals of the three batches were also confirmed by polarized light microscopy on the isolated products (data not shown). The observations under microscope were also confirmed by the particle size distribution data where the D50 of Batch 1 is less than half of what were in Batch 2 and Batch 3, and much wider particle size distribution was observed in Batch 1 (data not shown).

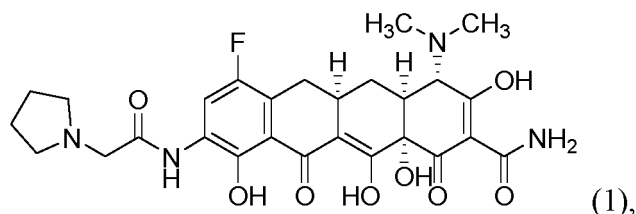
[0213] TGA and DSC analyses of the three batches were substantially identical.

[0214] The teachings of all patents, published applications and references cited herein are incorporated by reference in their entirety.

[0215] While this invention has been particularly shown and described with references to example embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the invention encompassed by the appended claims.

Claims

1. A crystalline form of the bis-HCl salt of a compound represented by Structural Formula 1:



wherein

the crystalline form is Form I and is **characterized by** at least three x-ray powder diffraction peaks at 2θ angles selected from 7.22° , 7.80° , 10.41° , and 11.11° , and wherein the crystalline form is a hemi-ethanolate solvate.

2. The crystalline form of claim 1, wherein the crystalline form is **characterized by**:

- (a) at least four x-ray powder diffraction peaks at 2θ angles selected from 7.22° , 7.80° , 8.19° , 10.41° , and 11.11° ;
- (b) at least five x-ray powder diffraction peaks at 2θ angles selected from 7.22° , 7.80° , 8.19° , 10.41° , 11.11° , 15.00° , 16.47° and 20.44° ;
- (c) x-ray powder diffraction peaks at 2θ angles of 7.22° , 7.80° , 8.19° , 10.41° , and 11.11° ;
- (d) x-ray powder diffraction peaks at 2θ angles selected from 7.22° , 7.80° , 8.19° , 10.41° , 11.11° , 15.00° , 16.47° and 20.44° ; or
- (e) x-ray powder diffraction peaks at 2θ angles of 7.22° , 7.80° , 8.19° , 10.41° , 11.11° , 12.17° , 13.52° , 15.00° , 15.70° , 16.47° , 19.96° , and 20.44° .

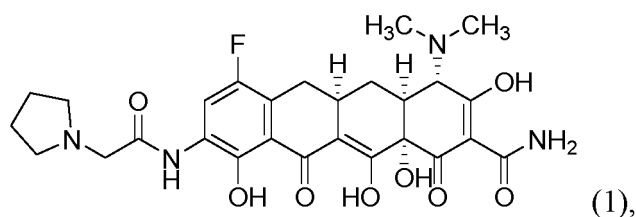
3. The crystalline form of claim 1 or claim 2, wherein the crystalline form is **characterized by** an x-ray powder diffraction pattern substantially in accordance with that depicted in Figure 5.

4. The crystalline form of any one of claims 1-3, wherein the compound of Structural Formula 1 is in the form of a co-solvate.

5. The crystalline form of claim 4, wherein the co-solvate is:

- (a) a water and ethanol solvate;
- (b) a dihydrate and hemi-ethanolate co-solvate; or
- (c) a variable co-solvate.

6. A crystalline form of the bis-HCl salt of a compound represented by Structural Formula 1:



wherein

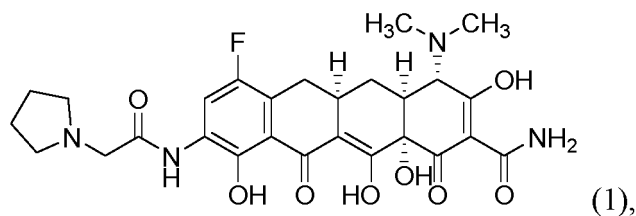
the crystalline form is Form J and is **characterized by** at least three x-ray powder diffraction peaks at 2θ angles selected from 7.02° , 7.80° , 22.13° , and 23.22° .

7. The crystalline form of claim 6, wherein the crystalline form is **characterized by**:

- (a) at least four x-ray powder diffraction peaks at 2θ angles selected from 7.02° , 7.80° , 10.25° , 22.13° , and 23.22° ;
- (b) at least five x-ray powder diffraction peaks at 2θ angles selected from 7.02° , 7.80° , 10.25° , 11.00° , 13.29° , 13.60° , 14.98° , 21.92° , 22.13° , 23.22° , 24.02° and 25.28° ;
- (c) x-ray powder diffraction peaks at 2θ angles of 7.02° , 7.80° , 10.25° , 22.13° , and 23.22° ;
- (d) x-ray powder diffraction peaks at 2θ angles of 7.02° , 7.80° , 10.25° , 11.00° , 13.29° , 13.60° , 14.98° , 21.92° , 22.13° , 23.22° , 24.02° and 25.28° ; or
- (e) x-ray powder diffraction peaks at 2θ angles of 7.02° , 7.80° , 10.25° , 11.00° , 11.85° , 13.29° , 13.60° , 14.98° , 15.27° , 16.21° , 16.39° , 17.04° , 20.10° , 21.53° , 21.92° , 22.13° , 22.52° , 23.22° , 24.02° , 24.41° , 25.28° , 26.08° , 26.35° , 26.78° , and 27.90° .

8. The crystalline form of claim 6 or claim 7, wherein the crystalline form is **characterized by** an x-ray powder diffraction pattern substantially in accordance with that depicted in FIG. 7.

9. A composition, comprising particles of one or more crystalline forms of a compound represented by the bis HCl salt of Structural Formula 1:



wherein the one or more crystalline forms are selected from:

the particles are of crystalline Form I **characterized by** at least three x-ray powder diffraction peaks at 2θ angles selected from 7.22° , 7.80° , 10.41° , and 11.11° , wherein the crystalline Form I is a hemi-ethanolate solvate; and the particles are of crystalline form Form J **characterized by** at least three x-ray powder diffraction peaks at 2θ angles selected from 7.02° , 7.80° , 22.13° , and 23.22° .

10. The composition of claim 9, wherein the composition comprises particles of crystalline Form I and crystalline form J.

11. The composition of claim 10, wherein the weight percent of crystalline Form J in the composition is 25% or less.

12. A pharmaceutical composition, comprising the compound of claim 1 and a pharmaceutically acceptable carrier.

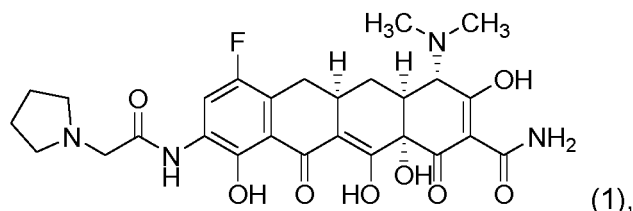
13. The composition of claim 12 for use in a method for treating or preventing an infection caused by bacteria, the method comprising administering to a subject in need thereof a therapeutically or prophylactically effective amount of the composition of claim 12.

14. The composition for use according to claim 13, wherein the infection is caused by a Gram-positive bacterium.

15. The composition for use according to claim 13, wherein the infection is caused by a Gram-negative bacterium.

Patentansprüche

1. Kristalline Form des Bis-HCl-Salzes einer durch Strukturformel 1 wiedergegebenen Verbindung:



wobei

die kristalline Form Form I ist und durch mindestens drei Röntgenpulverbeugungspeaks bei 2 θ -Winkeln, die aus 7,22°, 7,80°, 10,41° und 11,11° ausgewählt sind, gekennzeichnet ist und wobei es sich bei der kristallinen Form um ein Hemiethanolat-Solvat handelt.

2. Kristalline Form nach Anspruch 1, wobei die kristalline Form **gekennzeichnet ist durch:**

- (a) mindestens vier Röntgenpulverbeugungspeaks bei 2 θ -Winkeln, die aus 7,22°, 7,80°, 8,19°, 10,41° und 11,11° ausgewählt sind;
- (b) mindestens fünf Röntgenpulverbeugungspeaks bei 2 θ -Winkeln, die aus 7,22°, 7,80°, 8,19°, 10,41°, 11,11°, 15,00°, 16,47° und 20,44° ausgewählt sind;
- (c) Röntgenpulverbeugungspeaks bei 2 θ -Winkeln von 7,22°, 7,80°, 8,19°, 10,41° und 11,11°;
- (d) Röntgenpulverbeugungspeaks bei 2 θ -Winkeln, die aus 7,22°, 7,80°, 8,19°, 10,41°, 11,11°, 15,00°, 16,47° und 20,44° ausgewählt sind; oder
- (e) Röntgenpulverbeugungspeaks bei 2 θ -Winkeln von 7,22°, 7,80°, 8,19°, 10,41°, 11,11°, 12,17°, 13,52°, 15,00°, 15,70°, 16,47°, 19,96° und 20,44°.

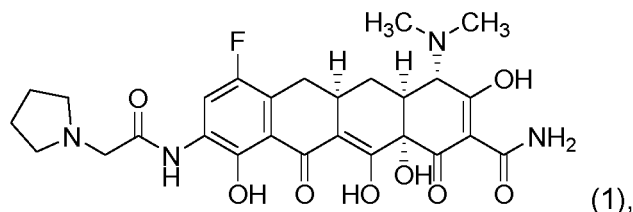
3. Kristalline Form nach Anspruch 1 oder Anspruch 2, wobei die kristalline Form durch ein Röntgenpulverbeugungsmuster gekennzeichnet ist, das weitgehend dem in Figur 5 dargestellten Muster entspricht.

4. Kristalline Form nach einem der Ansprüche 1-3, wobei die Verbindung der Strukturformel 1 in Form eines Cosolvats vorliegt.

5. Kristalline Form nach Anspruch 4, wobei es sich bei dem Cosolvat um:

- (a) ein Wasser- und Ethanol-Solvat;
- (b) ein Dihydrat- und Hemiethanolat-Cosolvat oder
- (c) ein variables Cosolvat handelt.

6. Kristalline Form des Bis-HCl-Salzes einer durch Strukturformel 1 wiedergegebenen Verbindung:



wobei

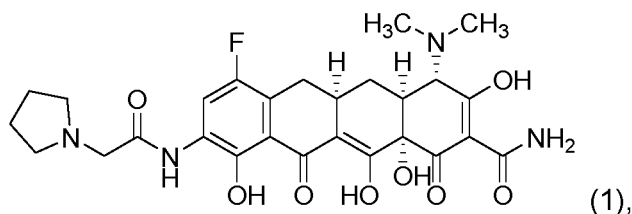
die kristalline Form Form J ist und durch mindestens drei Röntgenpulverbeugungspeaks bei 2 θ -Winkeln, die aus 7,02°, 7,80°, 22,13° und 23,22° ausgewählt sind, gekennzeichnet ist.

7. Kristalline Form nach Anspruch 6, wobei die kristalline Form **gekennzeichnet ist durch**:

- (a) mindestens vier Röntgenpulverbeugungspeaks bei 2θ -Winkeln, die aus $7,02^\circ$, $7,80^\circ$, $10,25^\circ$, $22,13^\circ$ und $23,22^\circ$ ausgewählt sind;
- (b) mindestens fünf Röntgenpulverbeugungspeaks bei 2θ -Winkeln, die aus $7,02^\circ$, $7,80^\circ$, $10,25^\circ$, $11,00^\circ$, $13,29^\circ$, $13,60^\circ$, $14,98^\circ$, $21,92^\circ$, $22,13^\circ$, $23,22^\circ$, $24,02^\circ$ und $25,28^\circ$ ausgewählt sind;
- (c) Röntgenpulverbeugungspeaks bei 2θ -Winkeln von $7,02^\circ$, $7,80^\circ$, $10,25^\circ$, $22,13^\circ$ und $23,22^\circ$;
- (d) Röntgenpulverbeugungspeaks bei 2θ -Winkeln von $7,02^\circ$, $7,80^\circ$, $10,25^\circ$, $11,00^\circ$, $13,29^\circ$, $13,60^\circ$, $14,98^\circ$, $21,92^\circ$, $22,13^\circ$, $23,22^\circ$, $24,02^\circ$ und $25,28^\circ$; oder
- (e) Röntgenpulverbeugungspeaks bei 2θ -Winkeln von $7,02^\circ$, $7,80^\circ$, $10,25^\circ$, $11,00^\circ$, $11,85^\circ$, $13,29^\circ$, $13,60^\circ$, $14,98^\circ$, $15,27^\circ$, $16,21^\circ$, $16,39^\circ$, $17,04^\circ$, $20,10^\circ$, $21,53^\circ$, $21,92^\circ$, $22,13^\circ$, $22,52^\circ$, $23,22^\circ$, $24,02^\circ$, $24,41^\circ$, $25,28^\circ$, $26,08^\circ$, $26,35^\circ$, $26,78^\circ$ und $27,90^\circ$.

8. Kristalline Form nach Anspruch 6 oder Anspruch 7, wobei die kristalline Form durch ein Röntgenpulverbeugungsmuster gekennzeichnet ist, das weitgehend dem in FIG. 7 dargestellten Muster entspricht.

9. Zusammensetzung, umfassend Teilchen einer oder mehrerer kristalline Formen einer Verbindung, die durch das Bis-HCl-Salz von Strukturformel 1 wiedergegeben wird:



wobei die eine oder die mehreren kristallinen Formen ausgewählt sind aus:

- den Teilchen der kristallinen Form I, **gekennzeichnet durch** mindestens drei Röntgenpulverbeugungspeaks bei 2θ -Winkeln, die aus $7,22^\circ$, $7,80^\circ$, $10,41^\circ$ und $11,11^\circ$ ausgewählt sind, wobei es sich bei der kristallinen Form I um ein Hemiethanolat-Sulfat handelt; und
- den Teilchen der kristallinen Form J, **gekennzeichnet durch** mindestens drei Röntgenpulverbeugungspeaks bei 2θ -Winkeln, die aus $7,02^\circ$, $7,80^\circ$, $22,13^\circ$ und $23,22^\circ$ ausgewählt sind.

10. Zusammensetzung nach Anspruch 9, wobei die Zusammensetzungsteilchen der kristallinen Form I und der kristallinen Form J umfasst.

11. Zusammensetzung nach Anspruch 10, wobei der Gewichtsprozentanteil von kristalliner Form J in der Zusammensetzung 25 % oder weniger beträgt.

12. Pharmazeutische Zusammensetzung, umfassend die Verbindung nach Anspruch 1 und einen pharmazeutisch unbedenklichen Träger.

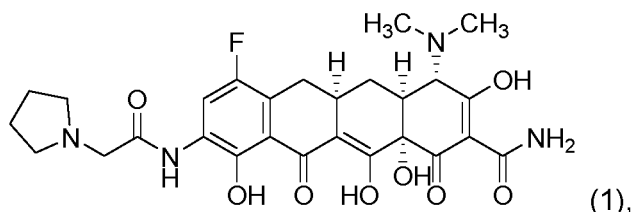
13. Zusammensetzung nach Anspruch 12 zur Verwendung bei einem Verfahren zur Behandlung oder Prävention einer durch Bakterien verursachte Infektionen, wobei das Verfahren das Verabreichen einer therapeutisch oder prophylaktisch wirksamen Menge der Zusammensetzung nach Anspruch 12 an ein Individuum, bei dem diesbezüglicher Bedarf besteht, umfasst.

14. Zusammensetzung zur Verwendung nach Anspruch 13, wobei die Infektion durch ein grampositives Bakterium verursacht wird.

15. Zusammensetzung zur Verwendung nach Anspruch 13, wobei die Infektion durch ein gramnegatives Bakterium verursacht wird.

Revendications

1. Forme cristalline du sel de bis-HCl d'un composé représenté par la formule structurale 1 :



la forme cristalline étant la formule I et étant **caractérisée par** au moins trois pics de diffraction des rayons X sur poudre à des angles 2θ choisis parmi $7,22^\circ$, $7,80^\circ$, $10,41^\circ$ et $11,11^\circ$, et la forme cristalline étant un solvate de type héli-éthanolate.

2. Forme cristalline selon la revendication 1, la forme cristalline étant **caractérisée par** :

- (a) au moins quatre pics de diffraction des rayons X sur poudre à des angles 2θ choisis parmi $7,22^\circ$, $7,80^\circ$, $8,19^\circ$, $10,41^\circ$ et $11,11^\circ$;
- (b) au moins cinq pics de diffraction des rayons X sur poudre à des angles 2θ choisis parmi $7,22^\circ$, $7,80^\circ$, $8,19^\circ$, $10,41^\circ$, $11,11^\circ$, $15,00^\circ$, $16,47^\circ$ et $20,44^\circ$;
- (c) des pics de diffraction des rayons X sur poudre à des angles 2θ de $7,22^\circ$, $7,80^\circ$, $8,19^\circ$, $10,41^\circ$ et $11,11^\circ$;
- (d) des pics de diffraction des rayons X sur poudre à des angles 2θ choisis parmi $7,22^\circ$, $7,80^\circ$, $8,19^\circ$, $10,41^\circ$, $11,11^\circ$, $15,00^\circ$, $16,47^\circ$ et $20,44^\circ$; ou
- (e) des pics de diffraction des rayons X sur poudre à des angles 2θ de $7,22^\circ$, $7,80^\circ$, $8,19^\circ$, $10,41^\circ$, $11,11^\circ$, $12,17^\circ$, $13,52^\circ$, $15,00^\circ$, $15,70^\circ$, $16,47^\circ$, $19,96^\circ$ et $20,44^\circ$.

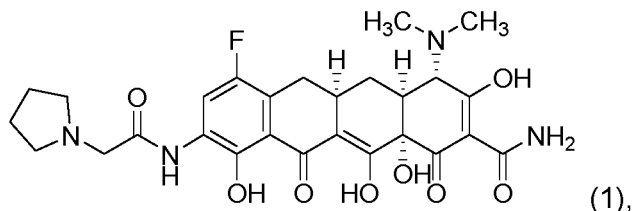
3. Forme cristalline selon la revendication 1 ou claim 2, la forme cristalline étant **caractérisée par** un diagramme de diffraction des rayons X sur poudre sensiblement conforme à celui représenté dans la figure 5.

4. Forme cristalline selon l'une quelconque des revendications 1 à 3, le composé de formule structurale 1 étant sous la forme d'un co-solvate.

5. Forme cristalline selon la revendication 4, le co-solvate étant :

- (a) un solvate avec l'eau et l'éthanol ;
- (b) un co-solvate de type dihydrate et héli-éthanolate ; ou
- (c) un co-solvate variable.

6. Forme cristalline du sel de bis-HCl d'un composé représenté par la formule structurale 1 :



la forme cristalline étant la forme J et étant **caractérisée par** au moins trois pics de diffraction des rayons X sur poudre à des angles 2θ choisis parmi $7,02^\circ$, $7,80^\circ$, $22,13^\circ$ et $23,22^\circ$.

7. Forme cristalline selon la revendication 6, la forme cristalline étant **caractérisée par** :

- (a) au moins quatre pics de diffraction des rayons X sur poudre à des angles 2θ choisis parmi $7,02^\circ$, $7,80^\circ$, $10,25^\circ$, $22,13^\circ$ et $23,22^\circ$;
- (b) au moins cinq pics de diffraction des rayons X sur poudre à des angles 2θ choisis parmi $7,02^\circ$, $7,80^\circ$, $10,25^\circ$,

11,00°, 13,29°, 13,60°, 14,98°, 21,92°, 22,13°, 23,22°, 24,02° et 25,28° ;

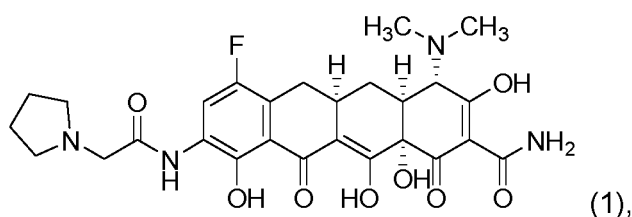
(c) des pics de diffraction des rayons X sur poudre à des angles 2θ de 7,02°, 7,80°, 10,25°, 22,13° et 23,22° ;

(d) des pics de diffraction des rayons X sur poudre à des angles 2θ de 7,02°, 7,80°, 10,25°, 11,00°, 13,29°, 13,60°, 14,98°, 21,92°, 22,13°, 23,22°, 24,02° et 25,28° ; ou

(e) des pics de diffraction des rayons X sur poudre à des angles 2θ de 7,02°, 7,80°, 10,25°, 11,00°, 11,85°, 13,29°, 13,60°, 14,98°, 15,27°, 16,21°, 16,39°, 17,04°, 20,10°, 21,53°, 21,92°, 22,13°, 22,52°, 23,22°, 24,02°, 24,41°, 25,28°, 26,08°, 26,35°, 26,78° et 27,90°.

8. Forme cristalline selon la revendication 6 ou la revendication 7, la forme cristalline étant **caractérisée par** un diagramme de diffraction des rayons X sur poudre sensiblement conforme à celui représenté dans la FIG. 7.

9. Composition, comprenant des particules d'une ou plusieurs formes cristallines d'un composé représenté par le sel de bis-HCl de formule structurale 1 :



la ou les formes cristallines étant choisies parmi :

les particules étant de forme cristalline I **caractérisée par** au moins trois pics de diffraction des rayons X sur poudre à des angles 2θ choisis parmi 7,22°, 7,80°, 10,41° et 11,11°, la forme cristalline I étant un solvate de type héli-éthanolate ; et

les particules étant de forme cristalline J **caractérisée par** au moins trois pics de diffraction des rayons X sur poudre à des angles 2θ choisis parmi 7,02°, 7,80°, 22,13° et 23,22°.

10. Composition selon la revendication 9, la composition comprenant des particules de forme cristalline I et de forme cristalline J.

11. Composition selon la revendication 10, le pourcentage en poids de la fonction cristalline J dans la composition étant de 25 % ou moins.

12. Composition pharmaceutique, comprenant le composé selon la revendication 1 et un support pharmaceutique acceptable.

13. Composition selon la revendication 12 pour une utilisation dans un procédé pour le traitement ou la prévention d'une infection causée par des bactéries, le procédé comprenant une administration à un sujet qui en a besoin d'une quantité efficace sur le plan thérapeutique ou sur le plan prophylactique de la composition selon la revendication 12.

14. Composition pour une utilisation selon la revendication 13, l'infection étant causée par une bactérie Gram-positive.

15. Composition pour une utilisation selon la revendication 13, l'infection étant causée par une bactérie Gram-négative.